Example Obtain an estimate of the domain of stability (DoS) for the origin for

$$\begin{pmatrix}\dot x\\ \dot y\end{pmatrix}=\begin{pmatrix}x(y^2-1)\\ y(x^2-1)\end{pmatrix}$$

Fixed points at $(0,0)$, $(\pm1,\pm1)$, $(\pm1,\mp1)$.

Linearization $\hat A=\begin{pmatrix}y^2-1 & 2xy\\ 2xy & x^2-1\end{pmatrix}$ gives

$\hat A(0,0)=\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}$, $\Rightarrow$ origin is a stable stellar node.

Other FPs are saddles. Symmetry w.r.t. $(x,y)\to(y,x)$, $x\to -x$, $y\to -y$.

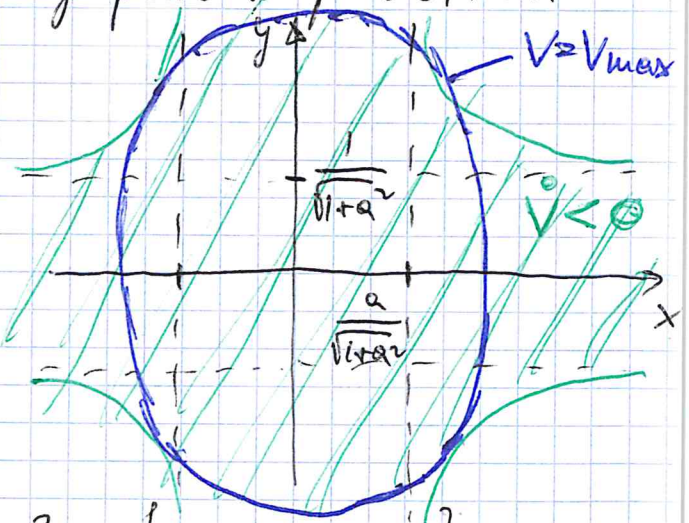
(1) Focus on the origin, try Lyapunov function

$V(\bar x)=x^2+a^2y^2$, $\Rightarrow$

$\dot V=2x\cdot x(y^2-1)+2a^2y\cdot y(x^2-1)$

$=-2(x^2+a^2y^2)+2(1+a^2)x^2y^2$.

We need to find D such that $\dot V(\bar x)<0$ for $\bar x\in D$.



D is given by

$$D=\left\{y^2<\frac{1}{1+a^2}\cdot\frac{x^2}{x^2-\frac{a^2}{1+a^2}}\right\}\cap\left\{|x|>\frac{a}{\sqrt{1+a^2}}\right\}\cup\left\{|x|<\frac{a}{\sqrt{1+a^2}}\right\}\ \text{(green)}$$

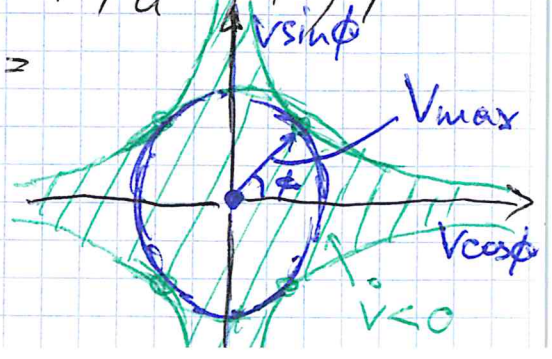
This is step (1).

(2) Need to find k: $C_k=\{\bar x: V(\bar x)<k\}\subseteq D$. Parametrize $V=$ const contour as $(x,y)=\sqrt{V}\left(\cos\phi,\frac{1}{a}\sin\phi\right)$, $\Rightarrow$

$\dot V=-2V+2(1+a^2)V\frac{1}{a^2}\sin^2\phi\cos^2\phi=$

$=-2V+2V\frac{1+a^2}{4a^2}\sin^2 2\phi<0$

for $V<\frac{4a^2}{(1+a^2)\sin^2 2\phi}$



Clearly, have $\dot{V} < 0$ guaranteed for

$0 < V < V_{max} = \frac{4a^2}{1+a^2}$ – this is k and

$C_k = \left\{ \bar{x} : x^2 + a^2 y^2 < \frac{4a^2}{1+a^2} \right\} \subset D$

Step (2) done.

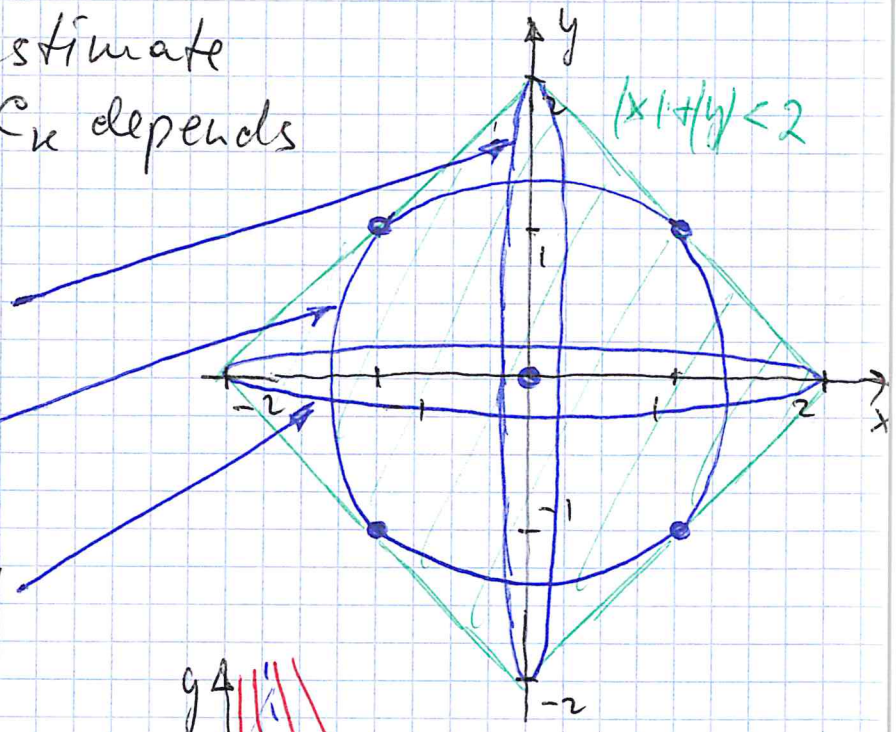
(3) Don't need to use La Salle here as within C_k have $\dot{V} < 0$, so can use second Lyapunov's theorem, which says that origin is asymptotically stable. $DoS(\bar{0}) = C_k$.

(4) Try to improve estimate of DoS: currently C_k depends on a, vary $a \Rightarrow$

$a \to 0$: $\left(\frac{x}{a}\right)^2 + y^2 \leq 4$

$a \to 1$: $x^2 + y^2 < 2$

$a \to \infty$: $x^2 + \left(\frac{y}{a-1}\right)^2 \leq 4$

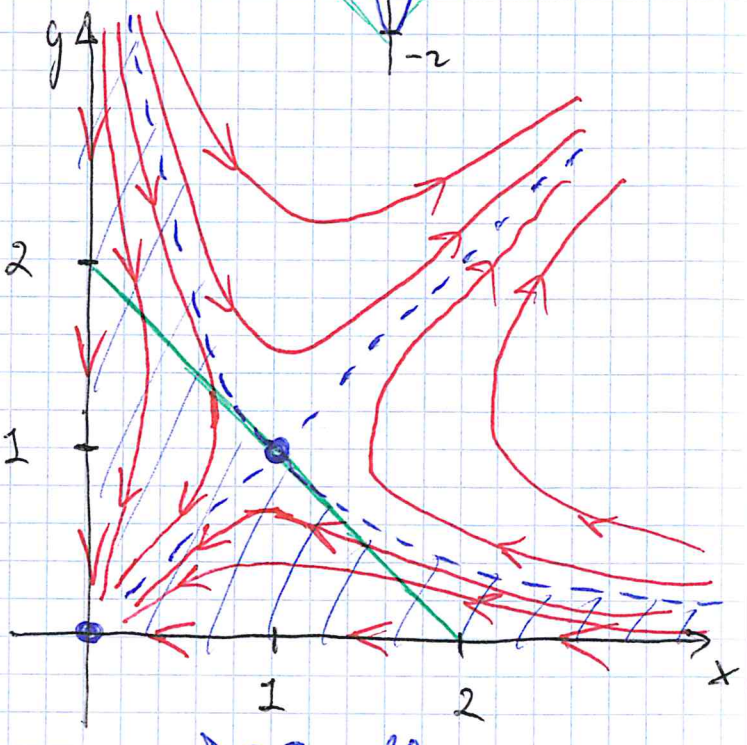


By varying a get different C_k, their union is $|x| + |y| < 2$.

This is the best DoS estimate obtained using this $V(\bar{x})$.

Is it the full DoS?

No! E.g. $x=0$, $y=0$ are always attracted to $\bar{0}$, even for $|y|>2$, $|x|>2$.



§3.3 Bounding functions

There is another use for functions decreasing along trajectories

Definition Bounding function

If V is a function with bounded contours and $\dot{V}(\bar{x}) < -\delta < 0$ for $V(\bar{x}) \geq M$, then all trajectories eventually enter and remain in $\{\bar{x} : V(\bar{x}) \leq M\}$. Such a function is called a bounding function.

Example Consider $\begin{cases} \dot{r} = r - r^3(1+k\sin\theta) - r^5, \\ \dot{\theta} = 1 \end{cases}$ $0 < k < 1$

Take $V = \frac{1}{2} r^2$, $\Rightarrow$

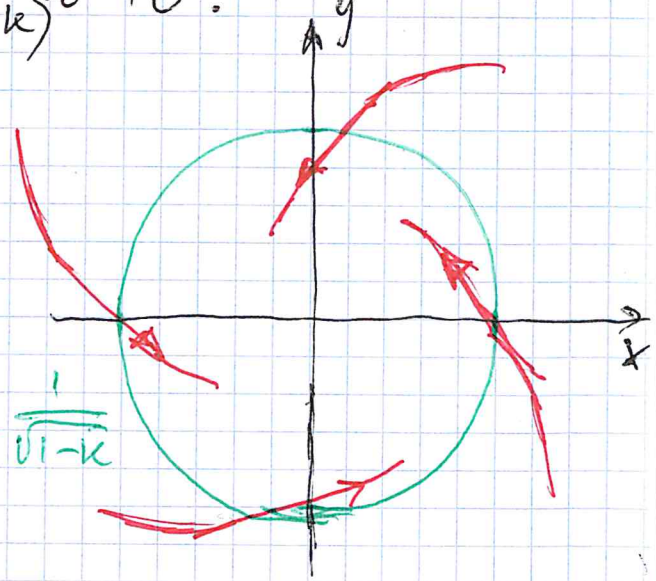
$\dot{V} = r\dot{r} = r^2\left[1 - r^2(1+k\sin\theta) - r^4\right]$.

Since $1 + k\sin\theta \geq 1 - k$ have $\dot{V} \leq r^2\left(1 - r^2(1-k) - r^4\right)$.

Then for $r^2 > \frac{1}{1-k}$ have

$\dot{V} < r^2\left[1 - 1 - \frac{1}{(1-k)^2}\right] = -\frac{r^2}{(1-k)^2} < 0$.

Thus, all trajectories enter the disc $r \leq \frac{1}{\sqrt{1-k}}$ and remain there.



What happens within this bounded domain, where we know ω-limit set exists?

Is it a fixed point? Or periodic orbit?

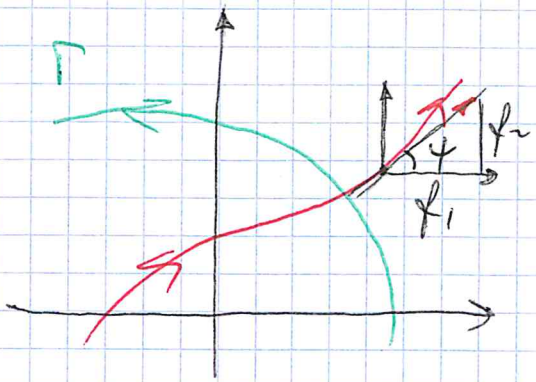
See next ...

§4. Periodic orbits in R^2: existence and stability

Let's first establish the techniques for confirming or excluding the existence of periodic orbits.

§4.1 Poincaré index

Start with $\begin{pmatrix}\dot x\\ \dot y\end{pmatrix} = \begin{pmatrix}f_1(x,y)\\ f_2(x,y)\end{pmatrix}$



Trajectory makes an angle $\psi = \tan^{-1}\frac{f_2}{f_1}$ with x-axis, if outside fixed points. Let Γ be simple closed curve (not necessarily a trajectory), go around it anticlockwise and measure the change of ψ along Γ. After one traverse ψ changes to its original value plus an integer multiple of 2π. This multiple is the Poincaré index $I(\Gamma)$ of Γ.

Explicit form

$$I(\Gamma) = \frac{1}{2\pi}\oint d\psi = \frac{1}{2\pi}\oint d\left(\tan^{-1}\frac{f_2}{f_1}\right) =$$

$$= \frac{1}{2}\oint \frac{1}{1+(f_2/f_1)^2}\cdot\frac{f_1 df_2 - f_2 df_1}{f_1^2} = \frac{1}{2\pi}\oint \frac{f_1 df_2 - f_2 df_1}{f_1^2+f_2^2}$$

Use any parametrization of Γ to compute $I(\Gamma)$, e.g. $df_2 = f_{2,x}\,dx + f_{2,y}\,dy$, use a circle for Γ, $y = R\sin\varphi$, $x = R\cos\varphi$, $\Rightarrow df_2 = R\,d\varphi\,(-f_{2,x}\sin\varphi + f_{2,y}\cos\varphi)$, etc. Note that at FP have $f_1, f_2 = 0$ and neither ψ nor $I(\Gamma)$ are defined.

Properties

(1) $I(\Gamma)$ is unaffected by a continuous deformation of Γ, as long as no FPs are crossed.

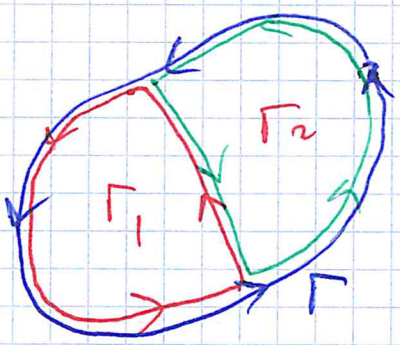
Indeed $I(\Gamma)$ takes only discrete, integer values, so changing $I(\Gamma)$ means it jumping by ± 1 or more. But deformation of Γ can be a series of tiny deformations, in which the integral for $I(\Gamma)$ would change only weakly. Which is compatible only if $I(\Gamma)$ stays constant in the process.

Note: can also see this by moving to complex plane, transforming $I(\Gamma)$ to a contour integral of an analytic function, which can have the contour deformed w/o crossing the pole.

(2) If Γ doesn't enclose FPs, $I(\Gamma)=0$ (because can deform to a tiny contour on which $\vec{F}=(f_1, f_2)$ is uniform and change of ψ is zero).

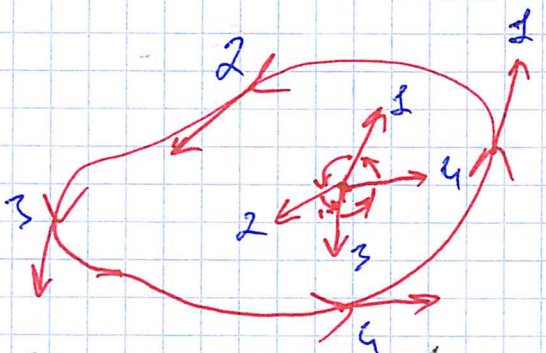


(3) $I(\Gamma)$ is additive, $I(\Gamma)=I(\Gamma_1)+I(\Gamma_2)$ if $\Gamma=\Gamma_1+\Gamma_2$ (joints cancel in the integral)



(4) Reversing time doesn't change $I(\Gamma)$, as ψ doesn't change when $f_{1,2} \to -f_{1,2}$.

(5) For a closed trajectory, e.g. a periodic orbit, $I(\Gamma)=+1$. This is since the field $\vec{F}$ is tangent to the orbit at every point.



(6) The Poincaré index of an isolated FP is the Poincaré index of any simple curve encircling this FP.

(7) For a curve encircling multiple FPs index